May 22nd, 2026

Direct and Large Eddy Simulations (DLES) 15

Delft, NL

Applicability of LES for Estimating Collision Statistics of Particles in Homogeneous Turbulence under Two-way Momentum Coupling

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ATMOSPHERIC TURBULENCE AND CLOUD DROPLET GROWTH

(smaller droplets)

DIFUSSIONAL GROWTH

SIZE GAP (20-60 μm)

St ~ 1

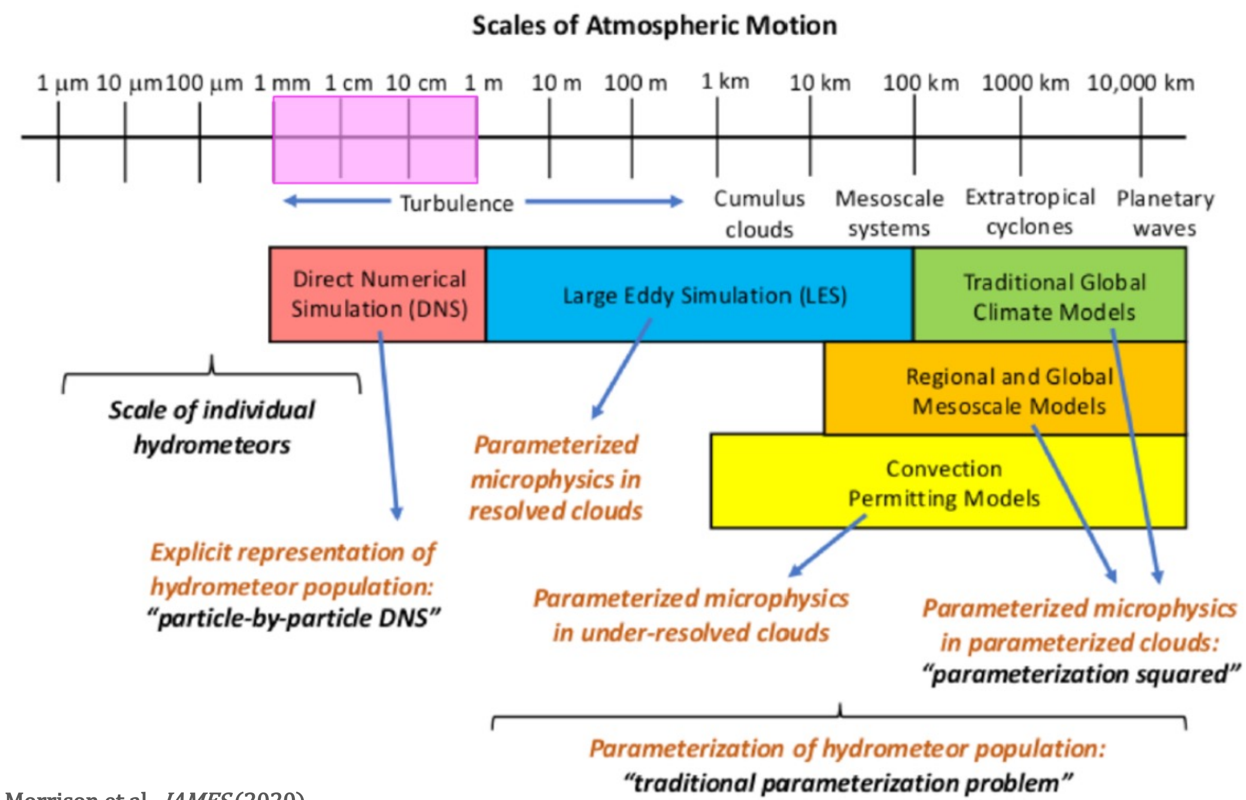
EFFECTS OF TURBULENCE

(larger droplets)

GROWTH DUE TO GRAVITATIONAL COLLISIONS

Grabowski & Wang, *Annu. Rev. Fluid Mech* (2013)

SCALES OF ATMOSPHERIC SIMULATIONS



Morrison et al., *JAMES* (2020)

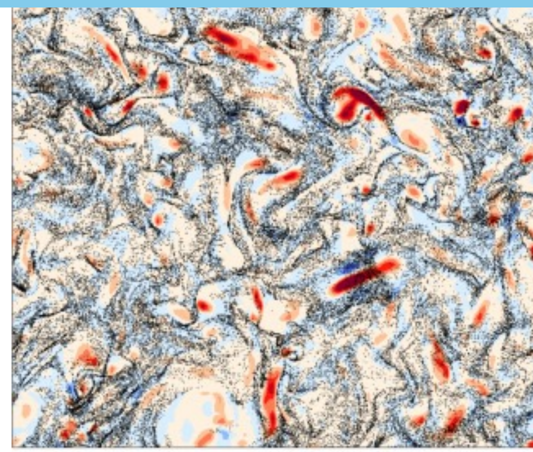
<u>FOCUS:</u>

Relatively small-scale simulations for fundamental research and estimating impact of turbulence on droplet collision rates

Perfect match for DNS!

DNS STUDIES

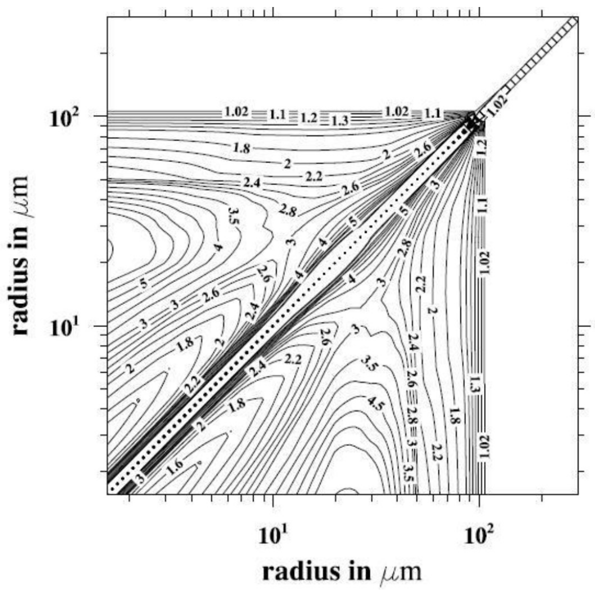
PREFERENTIAL CONCENTRATION



The overall enhancement factor of collection kernel by air turbulence.

Flow dissipation rate is 300 cm^2/s^3; r.m.s. velocity 202 cm/s.

COLLISION KERNELS



- Wang, Maxey, JFM (1993)
- Ayala, Grabowski, Wang, *JCP* (2007)
- Ayala, Rosa, Wang, Grabowski, *NJP* (2008)
- Wang, Rosa, Gao, He, Jin, *IJMF* (2009)
- Rosa, Parishani, Ayala, Wang, Grabowski, *NJP* (2013)
- Parishani, Ayala, Rosa, Wang, Grabowski, *Phys of Fluids* (2015)
- Rosa, Parishani, Ayala, Wang, *Phys. Of Fluids* (2015)
- Ababaei, Rosa, Pozorski, Wang, *JFM* (2021)

More recently, also incorporating two-way momentum coupling:

- Rosa, Pozorski, Wang, *IJMF* (2020)
- Rosa, Kopeć, Ababaei, Pozorski, *IJMF* (2022)

LIMITATIONS OF DNS – TOWARDS LES

Relatively recent scales of HIST DNS:

With particles:

- **R_λ = 531** on 2048^3 grid (Matsuda et al., *Phys. Rev. Fluids,* 2021)
- **R_λ = 597 on** 2048^3 grid (Ireland et al., *JFM,* 2016)

Flow-only:

- **R_λ = 650** on 8192^3 grid
- **R_λ = 1300** on $12\,288^3$ grid (Buaria et al., *NJP* 2019)

We observe atmospheric turbulence in clouds with R_λ ranging up to 10^4.

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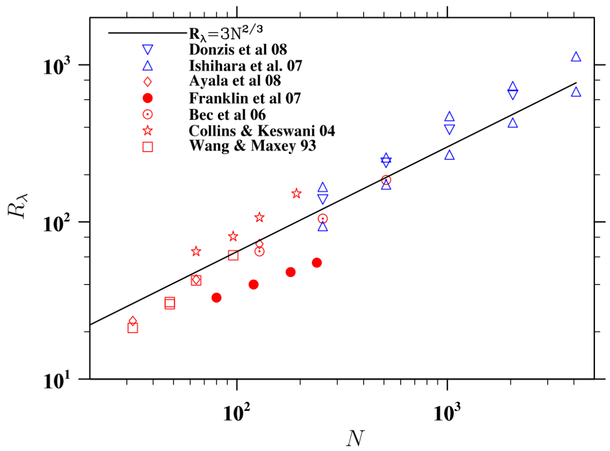
Fundamental scaling issue:

$R_\lambda \propto N^{2/3}$, while node count – N^3



Wang et al., IJMF, 2009, Figure 1

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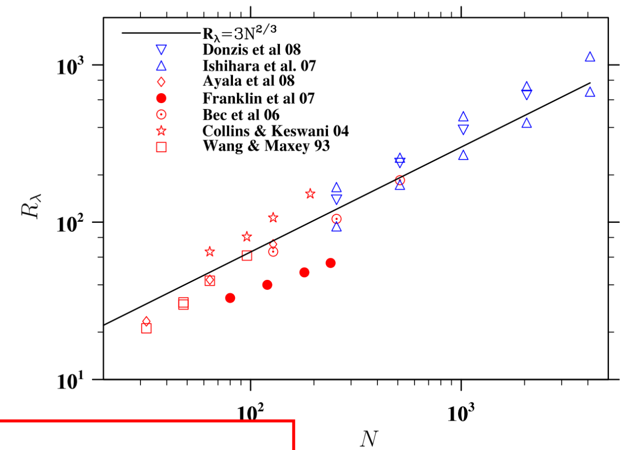


Wang et al., IJMF, 2009, Figure 1

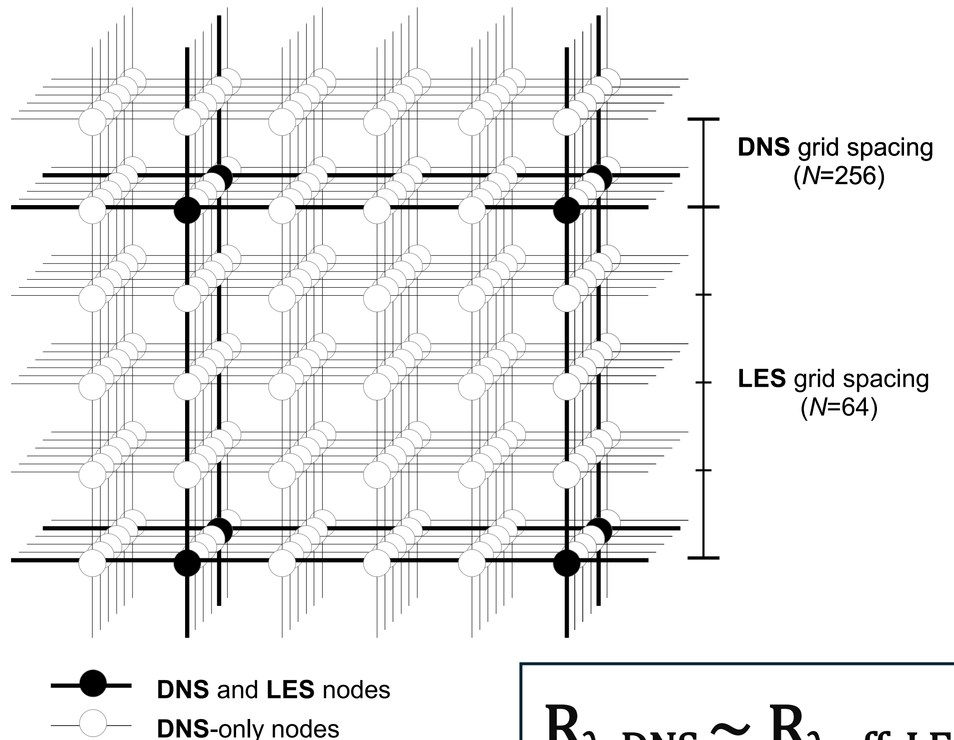
Using DNS to obtain larger R_λ is not feasible due to scaling issues!

TOWARDS LES

Initial studies on the feasibility of LES in these small-scale studies (importance of accuracy):

- Yang, He, Wang, *JoT* (2008)
- Jin, He, Wang, *Phys. of Fluids* (2010)
- Rosa, Pozorski, *JoT* (2017)

Limited to one-way momentum coupling between fluid and particles.



$$R_{\lambda,\,\mathrm{DNS}} \sim R_{\lambda\text{-eff},\,\mathrm{LES}}$$

OBJECTIVES OF THE STUDY

Establish what is the impact of using LES instead of DNS on the accuracy of estimating collision statistics (RDF, RRV, collision kernel) in simulations under two-way momentum coupling

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NOTE: Results from DNS are considered a baseline

METHOD

FLUID FLOW SOLVER

DNS

- pseudo-spectral solver for incompressible N-S equations
- periodic box with uniform mesh grid ($N = 256$)
- deterministic forcing to maintain stationary HIT
- massively parallel implementation (MPI-based), 2D domain decomposition

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LES

- simple adaptation of DNS solver to include SGS model (for better comparability)
- Smagorinsky model with spectral-eddy viscosity (Chollet, Lesieur, *JAS*, 1981; Rosa, Pozorski, *JoT*, 2017)

$$\left\{ \frac{\partial}{\partial t} + [\nu + \nu_e(k|k_c)]k^2 \right\} \hat{\bar{\mathbf{U}}}(\mathbf{k},t) = \mathbf{P}(\mathbf{k})\mathcal{F}(\bar{\mathbf{U}} \times \bar{\boldsymbol{\omega}}) + \hat{\mathbf{f}}(\mathbf{k},t),$$

$$\nu_e(k|k_c) = \nu_e^+(k|k_c)\sqrt{\frac{E(k_c)}{k_c}}, \qquad \nu_e^+(k|k_c) = C_K^{-3/2}[0.441 + 15.2\exp(-3.03k_c/k)],$$

FLUID FLOW SOLVER

	DNS	LES*	Rosa et al. (2013) DNS
N	256	64	256
$\Delta t \cdot 10^4$	9.0	9.0	9.0
$\nu \cdot 10^3$	1.5	1.5	1.1
u'	0.871 ± 0.001	0.861 ± 0.001	0.868 ± 0.002
ϵ	0.212 ± 0.002	0.200 ± 0.001*	0.200 ± 0.003
$\eta \cdot 10^2$	1.125 ± 0.003	2.078 ± 0.007*	0.903 ± 0.037
$\tau_k \cdot 10^2$	8.436 ± 0.043	15.755 ± 0.122*	7.420 ± 0.620
L_s	1.462 ± 0.003	1.501 ± 0.004	1.496 ± 0.005
$\lambda \cdot 10$	2.851 ± 0.011	5.252 ± 0.014*	2.494 ± 0.015
T_e	3.616 ± 0.028	3.697 ± 0.025*	3.774 ± 0.045
$k_{\max}\eta$	1.423 ± 0.004	—	1.143 ± 0.005
R_λ	165.91 ± 0.51	165.01 ± 0.11*	196.87 ± 0.78

COUPLING OF FLUID AND PARTICLES

FLUID

PARTICLES

- fluid drag (interpolated from nearest mesh nodes) applied according to Maxey & Riley (1983) – with or without gravity, no extra terms (history, etc.):

$$\begin{cases} \frac{d\mathbf{V}^{(k)}(t)}{dt} = -f(Re_p)\frac{\mathbf{V}^{(k)}(t)-\mathbf{U}(\mathbf{Y}^{(k)}(t),t)}{\tau_p} + \mathbf{g}, \\ \frac{d\mathbf{Y}^{(k)}(t)}{dt} = \mathbf{V}^{(k)}(t), \end{cases}$$

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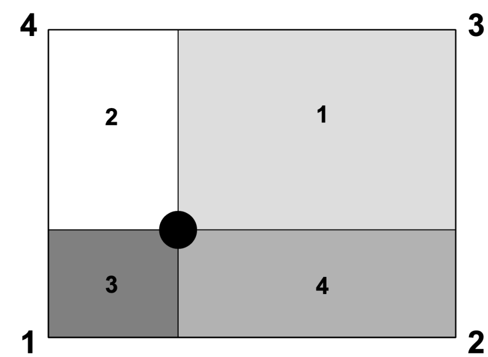
PARTICLES ← FLUID

- particle drag distributed to nearest nodes

(Bosse et al., *Phys. of Fluids* 2026; Rosa et al., *IJMF*, 2022):

$$\mathbf{f}^{(p)}(\mathbf{x},t) = -\frac{M}{\rho_p}\sum_{k=1}^{N_{\text{part}}} m_p^{(k)} \left(\frac{\mathbf{U}(\mathbf{Y}^{(k)}(t),t) - \mathbf{V}^{(k)}(t)}{\tau_p}\right) \delta\left(\mathbf{x} - \mathbf{Y}^{(k)}(t)\right),$$

- Used particle nearest neighbors (PNN) kernel (Garg et al., IJMF, 2009)



PARTICLE MASS LOADING AND SUPER-DROPLET PARAMETERIZATION

Under two-way momentum coupling the number of particles matters for the physics of a system. Here, measured in terms of the mass loading:

$$\Phi_m = \Phi_V \frac{\rho_p}{\rho} = \frac{(4/3)\pi a^3 N_{\text{part}}}{L^3} \frac{\rho_p}{\rho},$$

Due to limitations in number of computational particles (here – up to 20M), when necessary, a super-droplet parameterization was used (one computational particle carries mass of M particles) – for small M does not significantly affect results (see Rosa et al., *IJMF,* 2022; also partially verified here for M < 8)

DNS

	20 μm		30 μm		40 μm		50 μm		60 μm	
$\mathbf{\Phi_m}$	N_{part}	M	N_{part}	M	N_{part}	M	N_{part}	M	N_{part}	M
0.01	10.789	1	3.197	1	1.349	1	0.691	1	0.400	1
0.05	53.945	5	15.984	2	6.743	1	3.453	1	1.998	1
0.10	107.891	10	31.968	4	13.486	2	6.905	1	3.996	1
0.50	539.453	50	159.838	20	67.432	10	34.525	4	19.980	2
1.00	1 078.906	100	319.676	40	134.863	20	69.050	8	39.959	4

NOTE ON GRAVITY

Simulations are performed with and without effects of gravity on particles:

- NG – isolate effects of turbulence
- WG – full physical picture

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- NG – isolate effects of turbulence
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<u>DISCLAIMER:</u> Questions on the accuracy of TWC simulations with gravity using this method (DNS):

- anisotropy of modulated turbulence (due to effects of particles moving in preferred direction)
- unknown effects (non-physical) of removing mean velocity component from the modulated turbulent flow

FOCUS ON RESULTS WITHOUT GRAVITY IN THIS PRESENTATION!

RESULTS

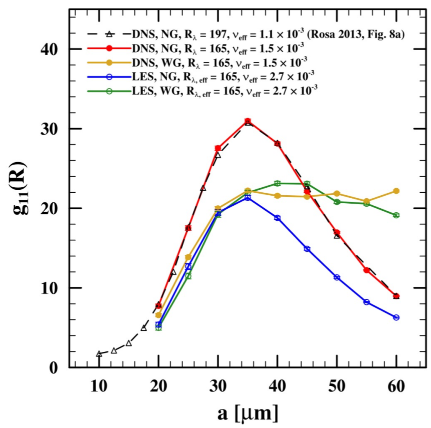
COLLISION STATISTICS OVERVIEW – RDF, RRV, COLLISION KERNELS

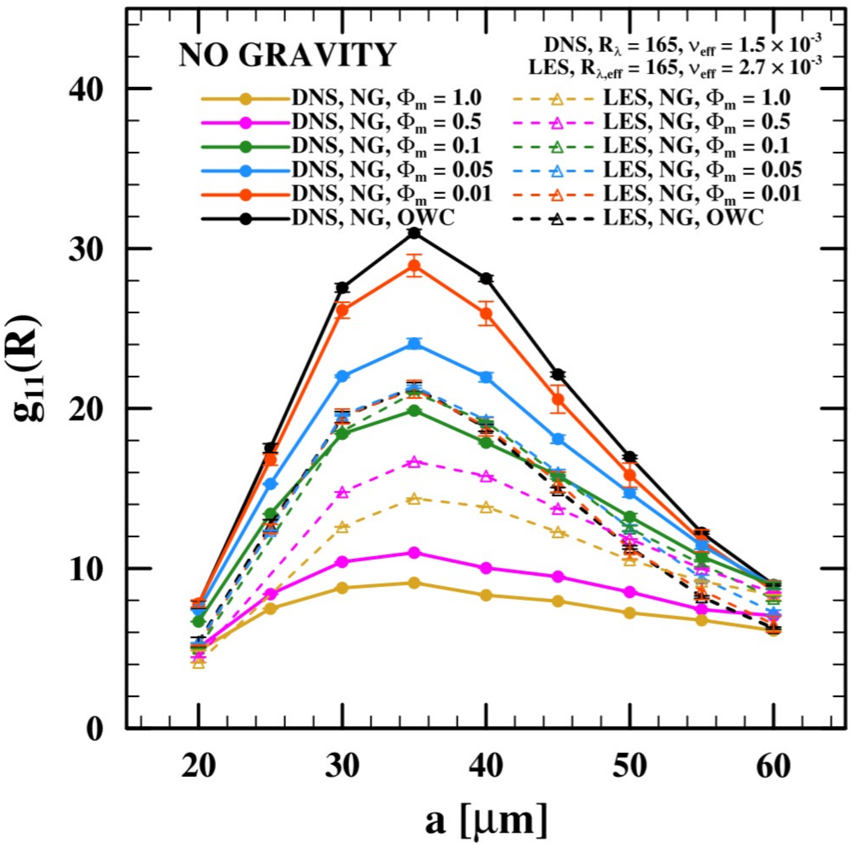
- **Radial Distribution Function (RDF) –** measure of preferential concentration of particles (at contact distance),
- **Radial Relative Velocity (RRV) –** collective measure of relative particle velocities and how they align with their separation (normalized – conventionally – by Kolmogorov velocity scale, at contact distance),
- **Collision Kernel –** ratio of the number of occurrences of particle collisions to the total number of particle pairs:
  - ***dynamic*** **–** explicitly counted
  - ***kinematic*** **–** calculated based on relation involving RDF and RRV:

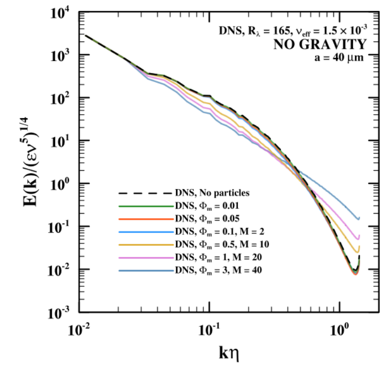
$$\Gamma_{11}^{K} = 2\pi R^2 \langle |w_{r,11}(r = R)| \rangle g_{11}(r = R),$$

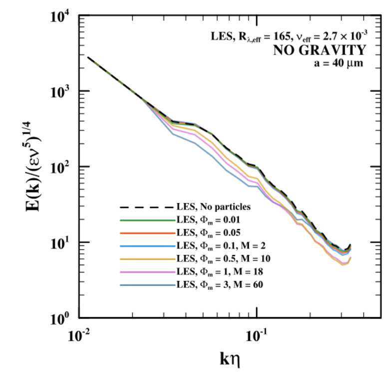
Both collision kernels were estimated for all presented results and - as expected (see Sundaram, Collins, *JFM*, 1997) **- they are equal within measurement uncertainties (necessary validation).**

RESULTS: RDF (1)

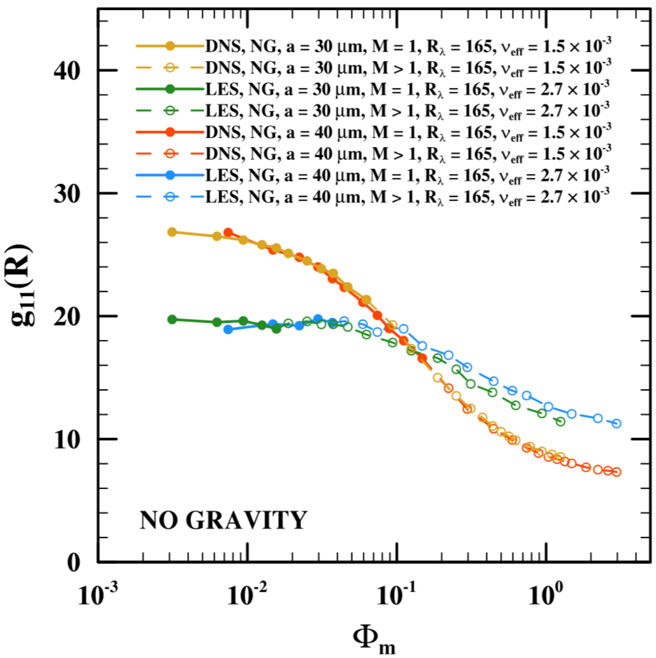






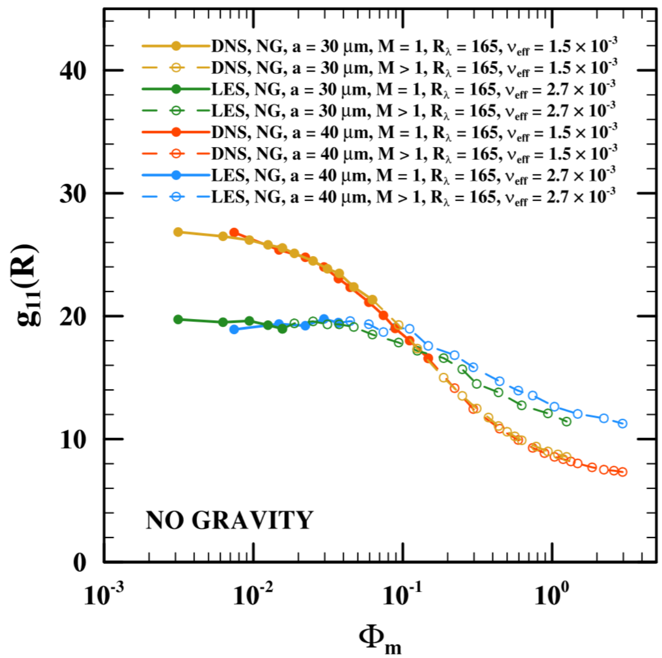


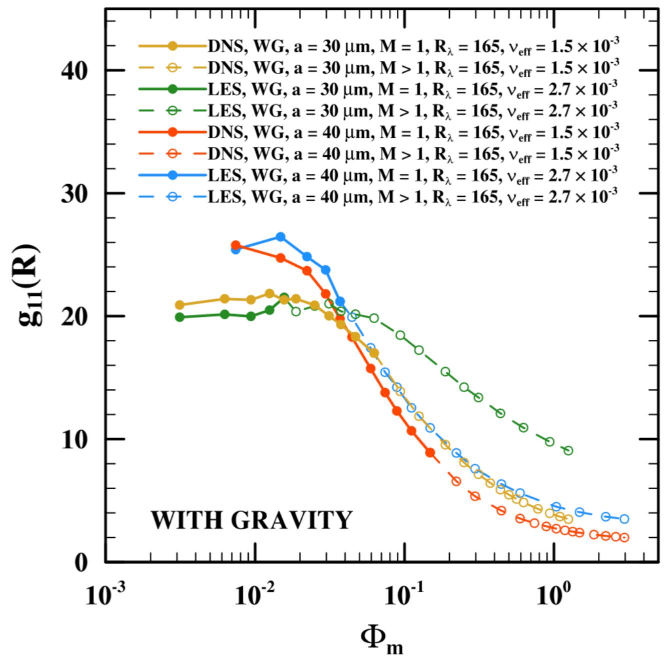
RESULTS: RDF (2)

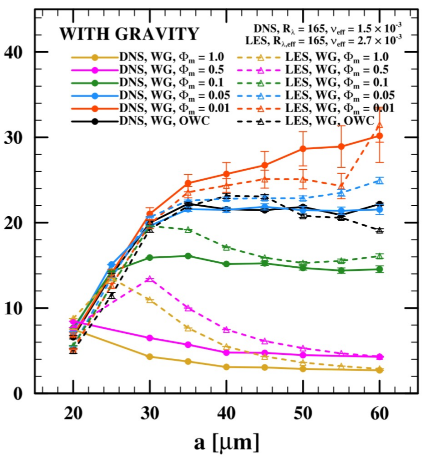


In agreement with results from OWC simulations, small-scale turbulence is important for preferential concentration of particles, therefore LES have problems to capture magnitude and changes in RDF based on system parameters (for mass loading < 0.1).

RESULTS: RDF (2) – ALSO WITH GRAVITY







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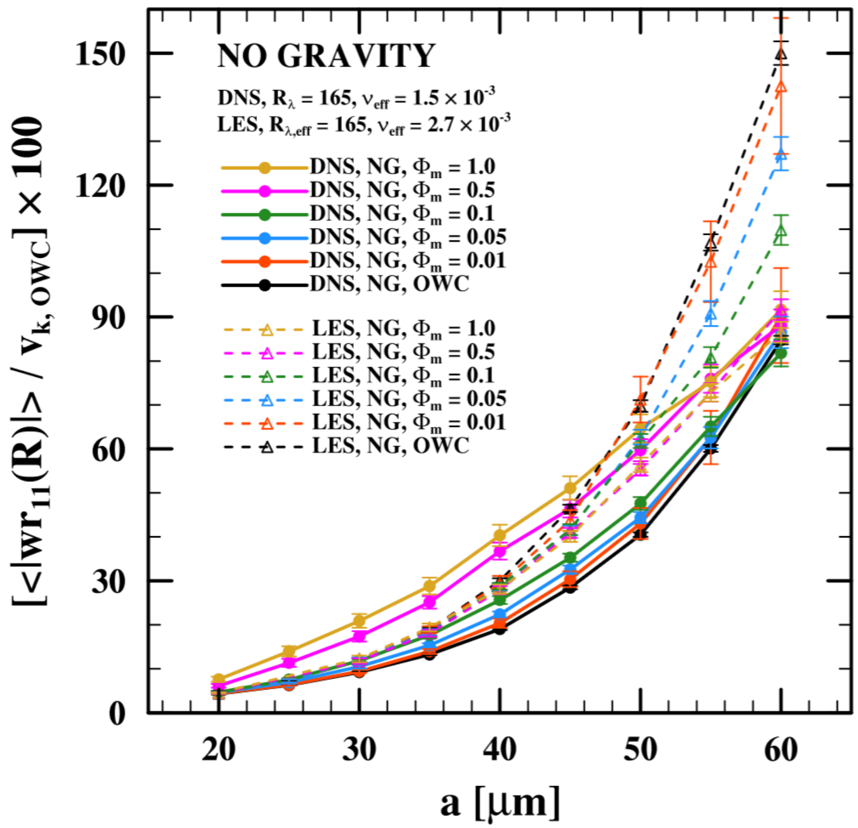
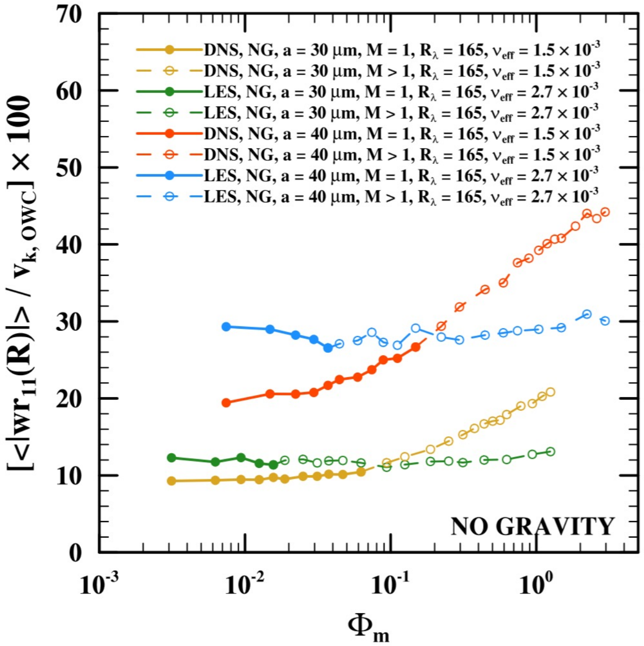
RESULTS: RRV (1)

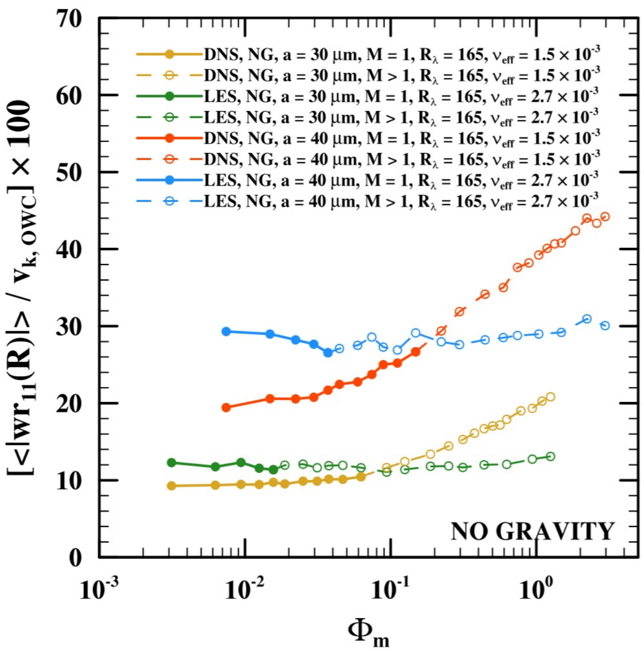


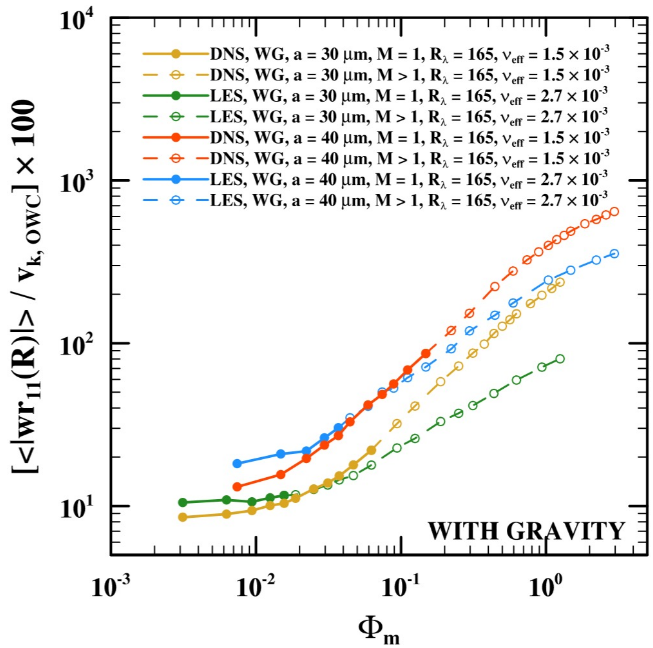
RESULTS: RRV (2)

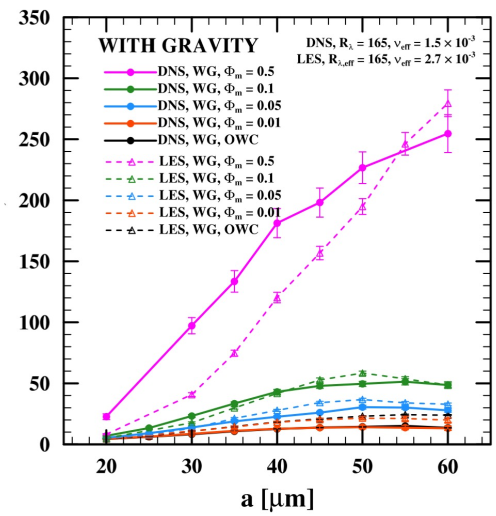


In agreement with results from OWC simulations, larger scales of turbulence affect RRV more, therefore we observe much closer alignment than in case of RDF. LES overestimates RVV for mass loadings < 0.1.

RESULTS: RRV (2) – ALSO WITH GRAVITY







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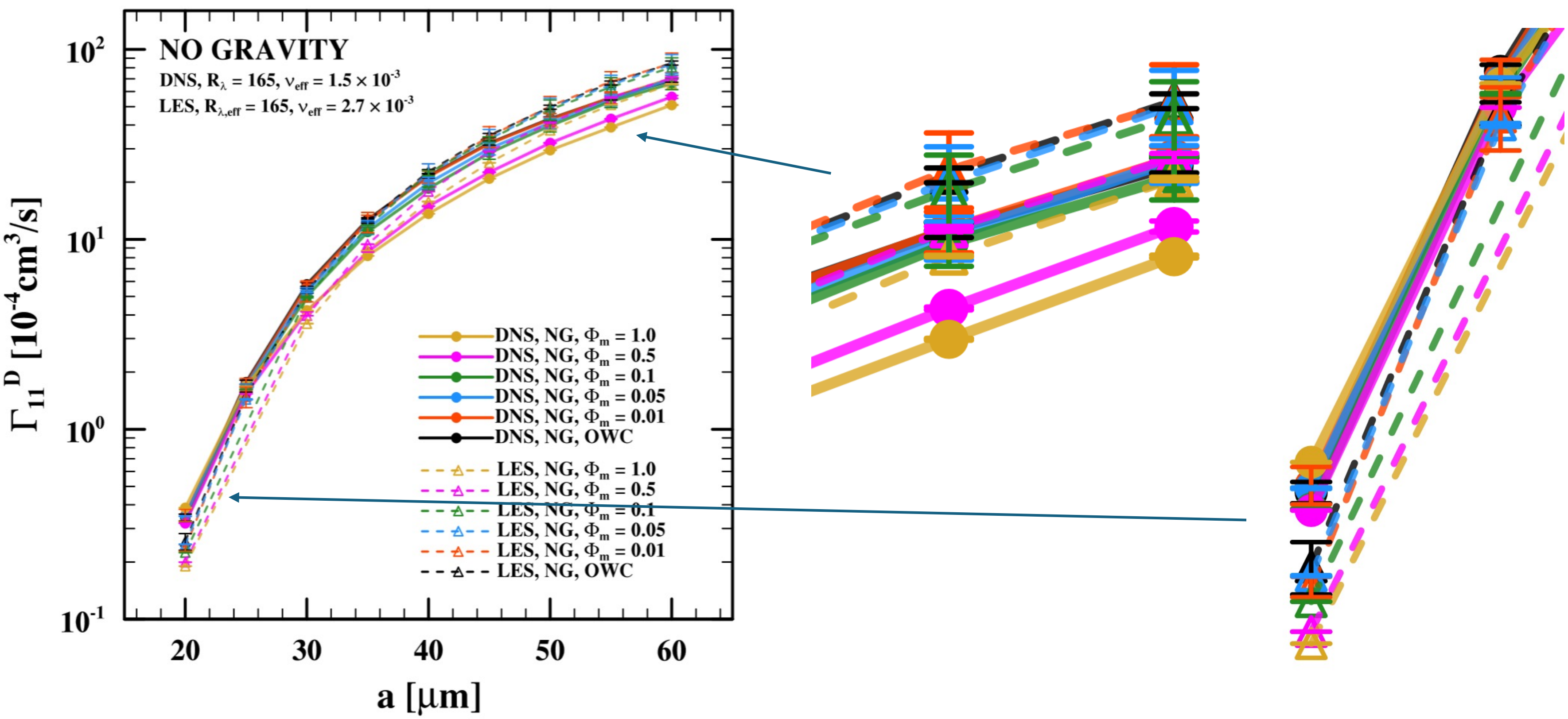
RESULTS: COLLISION KERNEL (1)



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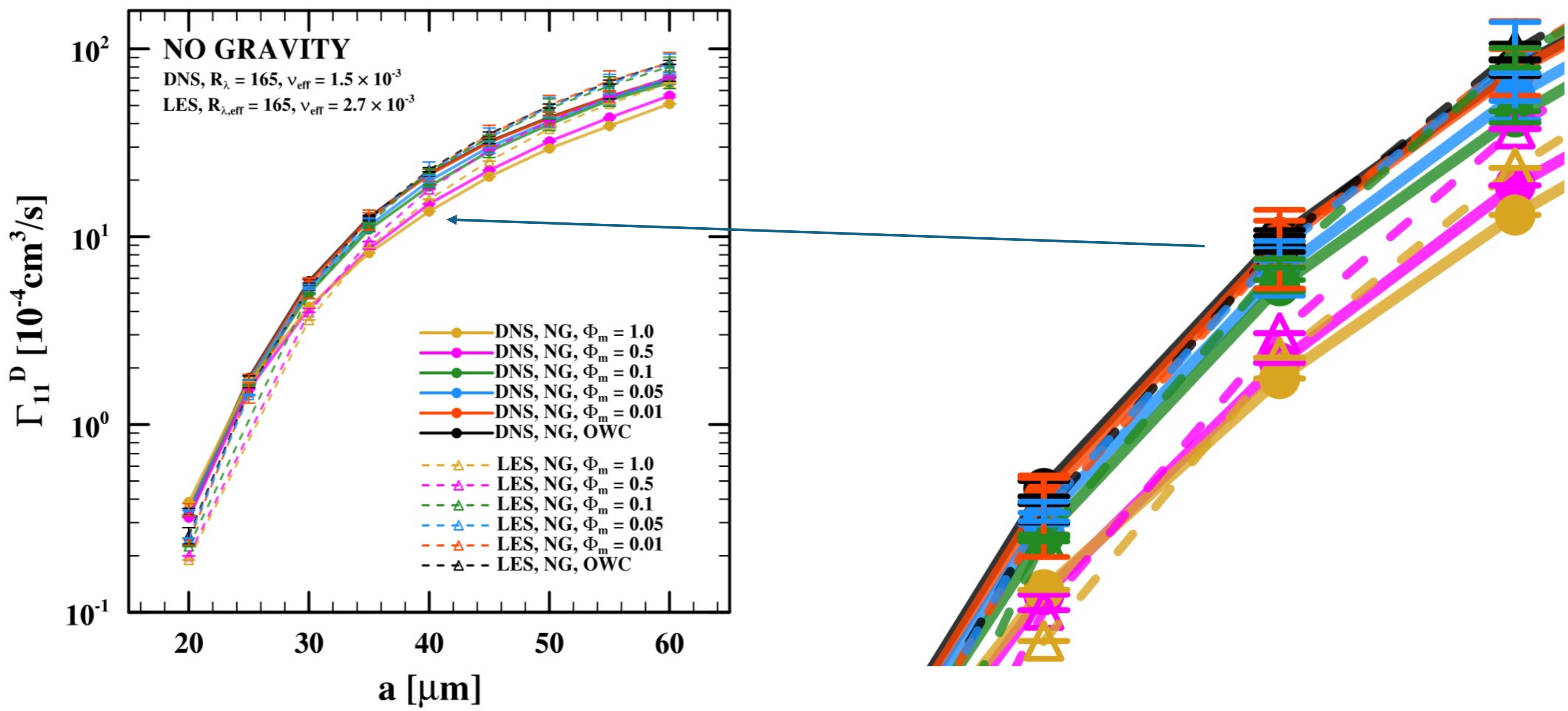
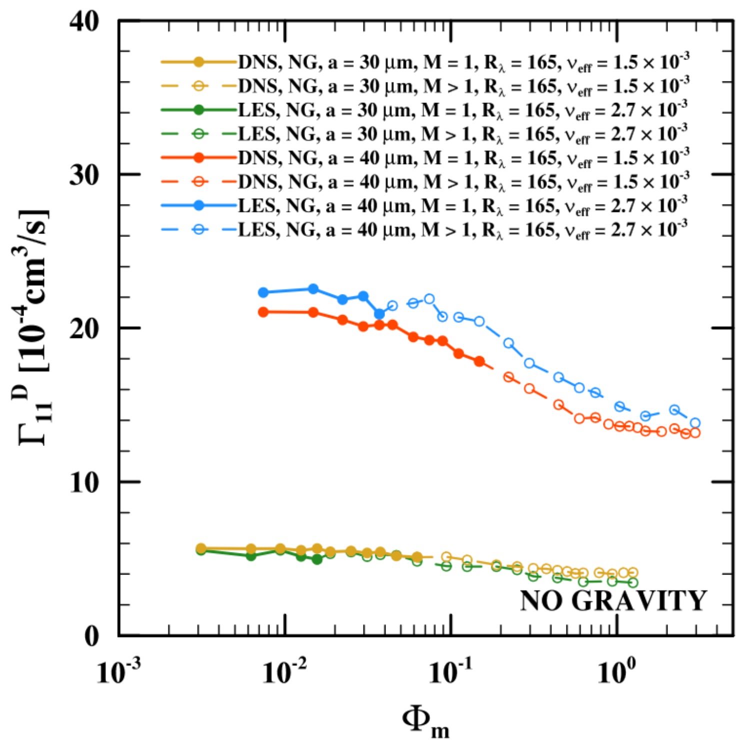
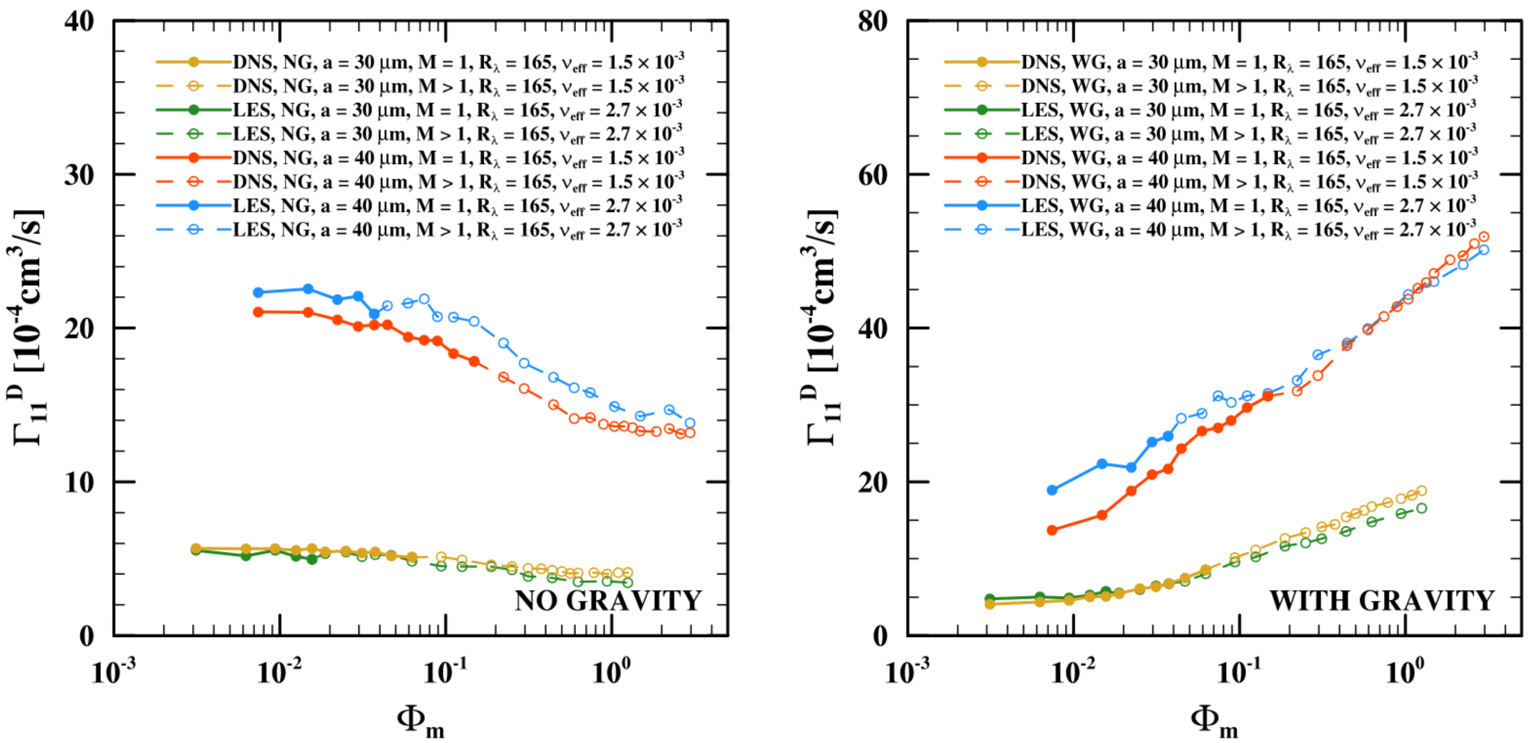


RESULTS: COLLISION KERNEL (3)



RESULTS: COLLISION KERNEL (3) – ALSO WITH GRAVITY



RESULTS: QUICK SUMMARY

- LES underestimates RDF for simulations with mass loading < 0.1 (most likely due to filtering of small-scale turbulence important for preferential concentration),
- LES overestimates RRV for simulations with mass loading < 0.1 (to a lesser degree, as larger turbulent scales that are resolved in LES are more impactful),
- LES statistics are less sensitive to changes in mass loading (less precise momentum transfer to sparser grid nodes; not modelled in SGS)
- for mass loadings > 0.1 these trends are reversed (more likely due to non-physical effects of super-droplet parameterization),
- both effects cancel each other to a significant degree(qualitatively and quantitatively), leading to a reasonable and accurate estimation of collision kernels (particles have higher velocities in direction aligned with gravity, therefore spend less time interacting with turbulent eddies – especially the small ones),
- current (tentative) results with gravity provide much more accurate match for all statistics considered.

FURTER WORK AND PERSPECTIVES

- broader analysis (more diverse simulation parameters, theoretical analysis) to explore whether cancellation of inaccuracies in RDF and RRV (also numerically) is accidental or an intrinsic property of this application,
- research and/or development of more particle-aware sub-grid scale models to reduce inaccuracies and eliminate found artifacts of turbulence parameterization (e.g. Balachandar et al., *Phys. Rev. Fluids*, 2024)
- adjustment of pseudo-spectral solver parallelization model to alleviate bottlenecks in particle-related computations (splitting particle population in one ”pencil” into several separate processes).

THANK YOU!

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